

30-Day Learning Roadmap

Generic Prompt vs. Context-Rich Prompt — A Side-by-Side Comparison

PROMPT ENGINEERING CASE STUDY

Topic used for both roadmaps: **Web Development (HTML, CSS, JavaScript fundamentals)**
Prepared to demonstrate how added context changes the quality of an AI-generated plan

PROMPT A — Roadmap Without Context

Prompt used: "Create a 30-day learning roadmap. Include weekly milestones, daily tasks, resources, projects, and final outcome. Make it practical and beginner-friendly." *(No details about the learner, topic, time, or goal were given — so a generic, assumed topic of "Web Development Basics" had to be picked.)*

Week 1 — Foundations

Milestone: Understand basic building blocks of websites (HTML & CSS)

Day	Task
1-2	Learn HTML structure: tags, elements, attributes
3-4	Learn CSS basics: selectors, colors, box model
5	Practice building a simple static page
6-7	Review + build a personal bio page

RESOURCES: freeCodeCamp HTML/CSS course, MDN Web Docs

PROJECT: A one-page personal profile site

Week 2 — Styling & Layout

Milestone: Be able to design responsive layouts

Day	Task
8-9	Flexbox basics
10-11	CSS Grid basics
12	Responsive design / media queries
13-14	Rebuild Week 1 project responsively

RESOURCES: CSS-Tricks guides, YouTube tutorials

PROJECT: Responsive landing page

Week 3 — JavaScript Basics

Milestone: Understand programming logic with JS

Day	Task
15-17	Variables, data types, functions
18-19	Loops and conditionals
20-21	DOM manipulation basics

RESOURCES: JavaScript.info, freeCodeCamp JS course

PROJECT: Interactive to-do list app

Week 4 — Integration & Portfolio

Milestone: Combine everything into a complete project

Day	Task
22-24	Build a multi-page mini website
25-27	Add interactivity with JS
28-29	Polish, debug, deploy (e.g., GitHub Pages)
30	Final review and reflection

RESOURCES: GitHub Pages docs, general tutorials

PROJECT: A small personal portfolio website

FINAL OUTCOME: A basic working knowledge of HTML, CSS, and JavaScript, with a simple personal portfolio website as proof of learning.

PROMPT B — Roadmap With Context

Context supplied: Current Situation: *Working Professional (full-time job)* | Current Skills: *Comfortable with Excel, no coding background* | Goal: *Build and launch a personal portfolio website to support a career switch into front-end development* | Available Time: *1.5 hrs/day on weekdays, 3 hrs/day on weekends* | Experience Level: *Beginner* | Preferred Learning Style: *Project-based, with short videos for new concepts*

Week 1 — HTML & CSS Foundations (≈11 hrs)

Milestone: Ship a static personal "About Me" page online — your first live URL		
Day	Task (sized to your time budget)	Why it matters for your goal
1 (1.5h)	Watch 1 short video on how websites work; set up VS Code + browser	Removes setup friction before coding starts
2 (1.5h)	HTML tags & structure — write your own bio page skeleton	Directly becomes Section 1 of your portfolio
3 (1.5h)	CSS basics: colors, fonts, spacing on your bio page	Makes the portfolio look intentional, not raw HTML
4 (1.5h)	Box model + positioning — fix layout issues on your page	Skill recruiters/hiring managers visually notice first
5 (1.5h)	Add a profile photo, links, simple nav bar	Makes the page feel like a real personal site
6 (3h, Sat)	Deploy the page live using GitHub Pages	You get a real URL to share in Week 1 — fast motivation win
7 (3h, Sun)	Polish + get feedback from 1 friend/peer	Builds the feedback habit you'll need as a developer

RESOURCES (MATCHED TO YOUR STYLE): Kevin Powell YouTube (CSS, short videos), MDN "Getting Started with HTML", GitHub Pages quickstart guide
PROJECT: Live "About Me" page with your real bio — usable in LinkedIn right away

Week 2 — Responsive Layout + Excel-to-Web Bridge (≈11 hrs)

Milestone: Build a responsive multi-section layout (you already think in "tables/grids" from Excel — Grid will click fast)		
Day	Task	Why it matters
8 (1.5h)	Flexbox — align a navbar and project cards	Used in 90% of real portfolio sites
9 (1.5h)	CSS Grid — note the parallel to Excel rows/columns	Familiar mental model speeds this up for you specifically
10 (1.5h)	Media queries — make site work on mobile	Recruiters often check sites on phones
11 (1.5h)	Add a "Skills" section styled as cards/grid	Translates your Excel/professional skills visually
12 (1.5h)	Add a "Career Switch Story" section	Tells your narrative — important for career-switchers
13 (3h, Sat)	Full responsive pass + cross-device testing	Catch issues before showing to anyone
14 (3h, Sun)	Mini-project: rebuild your Excel resume as an HTML table styled with CSS	Bridges your existing Excel skill directly into web skill

RESOURCES: Kevin Powell "Grid" playlist, web.dev responsive design guide
PROJECT: Responsive multi-section portfolio shell (About / Skills / Story)

Week 3 — JavaScript for Interactivity (≈11 hrs)

Milestone: Make your portfolio interactive — the signal that separates "designer" from "developer" candidates		
Day	Task	Why it matters
15 (1.5h)	JS basics: variables, functions (short video + practice)	Minimum needed before touching the DOM
16 (1.5h)	Conditionals & loops — small practice drills	Logic foundation for interview-style questions later
17 (1.5h)	DOM selection & event listeners	This is literally "click button → something happens"
18 (1.5h)	Build a dark/light mode toggle for your site	A small, resume-worthy interactive feature
19 (1.5h)	Build a simple contact form with validation	Recruiters expect a way to reach you
20 (3h, Sat)	Build a filterable "Projects" grid (filter by tag)	Shows you can handle dynamic data — a real hiring signal
21 (3h, Sun)	Debug + connect all JS features together	Integration practice mirrors real dev work

RESOURCES: JavaScript.info (read-along), "JS DOM Crash Course" video, freeCodeCamp JS basics for drills
PROJECT: Interactive portfolio with toggle, contact form, and filterable project grid

Week 4 — Launch & Career-Switch Packaging (≈11 hrs)

Milestone: A launched, shareable portfolio + the materials to start applying		
Day	Task	Why it matters
22 (1.5h)	Add 2 real project case studies to the "Projects" section	This is what hiring managers scan for first
23 (1.5h)	Write clear, simple copy for each section (no jargon)	Career-switchers are judged on clarity, not buzzwords
24 (1.5h)	Custom domain setup (optional) + SEO basics (title, meta tags)	Makes the site discoverable and professional
25 (1.5h)	Accessibility pass (alt text, color contrast, keyboard nav)	Increasingly checked in real front-end interviews
26 (1.5h)	Performance check + final bug fixes	Shows attention to detail — important with no coding history yet
27 (3h, Sat)	Record a 2-min Loom walkthrough of the site	Gives recruiters a fast way to "see" your thinking
28 (3h, Sun)	Update LinkedIn + resume to link the live portfolio	Converts the project into actual job-search material
29 (1.5h)	Get feedback from 2 people in tech (forums/communities)	External validation before going public
30 (1.5h)	Public launch post (LinkedIn/Twitter) + reflection	Starts building your visibility in the dev community

RESOURCES: web.dev accessibility basics, r/webdev & Frontend Mentor community for feedback
PROJECT: Final, launched, case-study-rich portfolio site tied to your career switch story
FINAL OUTCOME: A live, responsive, interactive portfolio website — built around your actual career-switch narrative — plus an updated LinkedIn/resume linking to it and direct community feedback, ready to use in real front-end job applications.

Dimension	Prompt A (No Context)	Prompt B (With Context)
Topic selection	Had to be guessed/assumed — no way to know what the learner actually wants	Anchored to a real, stated goal (career switch into front-end dev)
Daily task sizing	Generic "Day 1-2, Day 3-4" blocks with no time estimate	Each task sized to actual available hours (1.5h weekday / 3h weekend)
Relevance of resources	Generic, popular resources (freeCodeCamp, MDN) with no link to how the learner likes to learn	Matched to stated learning style (short videos + hands-on projects)
Use of prior skills	None — treats the learner as a blank slate	Explicitly bridges existing Excel skill into CSS Grid and resume-to-HTML project
Projects	Generic portfolio site with no connection to a real goal	Portfolio is the actual deliverable the learner needs for their job search
Final outcome	"Basic knowledge + a simple website" — vague, no next step	Concrete, real-world output: live portfolio + updated resume/LinkedIn + feedback loop, ready to apply for jobs
Motivation design	No early wins built in	Deliberate early win (live URL by Day 6) to sustain motivation

1. Which roadmap feels more personalized?

Prompt B is clearly more personalized. It doesn't just insert the learner's name into a template — it actively shapes *what* is taught, *how much*

2. Which roadmap would you actually follow?

Prompt B's roadmap is the one a real learner would be far more likely to stick with for all 30 days. The daily time blocks match a real schedule instead of assuming unlimited free time, the early "live URL by Day 6" win creates momentum, and the Excel-to-CSS-Grid bridge reduces the intimidation factor for someone with no coding background. Prompt A's roadmap is usable as a rough skeleton, but a learner would have to manually re-size every task and guess at relevance — extra friction that increases the chance of dropping off mid-month.

3. What role did context play in improving the result?

Context changed the roadmap from a **generic information dump** into a **decision-making tool**. Specifically, context allowed the plan to:

- **Right-size effort** — tasks fit real available hours instead of an assumed "free all day" learner
- **Prioritize relevance** — every resource and project ties back to the stated end goal rather than to "web dev in general"
- **Reuse existing strengths** — prior skills (Excel) were treated as an asset to build on, not ignored
- **Sequence for motivation** — context about experience level allowed early wins to be deliberately placed
- **Sharpen the final outcome** — the end state became a specific, usable deliverable (job-ready portfolio) instead of a vague skill claim

Bottom line: The instructions given to the AI were nearly identical in both prompts — the structure (milestones, daily tasks, resources, projects, outcome) was the same. The difference in quality came entirely from **context**. This confirms a core principle of effective prompting: *structure tells the AI what shape to produce; context tells it what to actually put inside that shape*. Without context, even a well-structured prompt produces a generic, average-case answer. With context, the same structure produces a plan that feels custom-built for one specific person.